S I M M Raton Mondol

Post-Doctoral Fellow · Information and Electronics Joint Research Institute

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Profile

Postdoctoral researcher with a Ph.D. in Electrical and Computer Engineering, specializing in biomedical informatics, speech processing, and AI/ML for healthcare. Experienced in applying ML to clinical and speech datasets, especially for Parkinson's disease. Skilled in building classification pipelines integrating acoustic features and ASR outputs. Passionate about developing scalable informatics tools for data-driven clinical outcomes.

Education

Ph.D. in Electrical and Computer Engineering *Sep. 2017 – Aug. 2024*
Inha University, Incheon, Korea *GPA:* ***4.38/4.50***

M.Eng. in Information and Communication Engineering *Sep. 2013 – Jun. 2016*
Chongqing University of Posts and Telecommunications, Chongqing, China *Score:* ***88%***

B.Sc. in Electronics and Telecommunication Engineering *May 2007 – Feb. 2012*
Daffodil International University, Dhaka, Bangladesh *GPA:* ***3.81/4.00***

Technical Skills

- **Programming Languages:** Python, MATLAB
- **Libraries & Frameworks:** PyTorch, Scikit-learn, TensorFlow, Pandas, Keras, NumPy, SciPy
- **Machine Learning Models:** SVM, RF, MLP, CNN, DNN
- **Speech/Clinical Tools:** Whisper, Wav2Vec 2.0, COVAREP, Praat, Parselmouth

Research Experience

Postdoctoral Fellow *Sep. 2024 – Present*
Information and Electronics Joint Research Institute, Inha University

- Advancing prior doctoral research by developing next-generation AI models for Parkinson's severity staging using enhanced acoustic-ASR feature fusion.
- Leading experiments on scalable, multimodal diagnostic pipelines that integrate speech features with clinical data for improved precision.

Graduate Research Assistant *Sep. 2017 – Aug. 2024*
Dept. of Electrical and Computer Engineering, Inha University

- Development of machine learning pipelines for multistage classification of Parkinson's disease using acoustic features and ASR outputs, including Whisper fine-tuning.
- Contributed to clinical applications of ML through personalized hearing aid fitting and speech recognition optimization for Korean cochlear implant patients.

Graduate Research Assistant *Sep. 2014 – Jun. 2016*
Dept. of Information and Communication Engineering, CQUPT, China

- Implemented impulse response-based acoustic echo cancellation systems using speech and random signals.
- Contributed to research on double-talk detection, and voice activity detection for acoustic echo cancellation.

Teaching & Academic Service

Teaching Assistant *Mar. 2022 – Aug. 2022*
College of Software Convergence, Inha University

International Student Ambassador *Mar. 2021 – Aug. 2024*
International Center, Inha University

Professional Experience

System Administrator *Apr. 2012 – Jul. 2013*
ICE Technologies Ltd., Dhaka, Bangladesh

Assistant System Engineer *Apr. 2011 – Mar. 2012*
Triangle Services Ltd., Dhaka, Bangladesh

Selected Publications

Journals

1. **Mondol, S.**, Kim, R., Lee, S. (2025). "Advanced optimization strategies for combining acoustic features and speech recognition error rates in multi-stage classification of Parkinson's disease severity." *Biomed. Eng. Lett., 15(3), 497.* (ISSN: 2093-985X, I.F. = 3.2) DOI: 10.1007/s13534-025-00465-9.
2. **Mondol, S.**, Kim, R., Lee, S. (2023). "Hybrid Machine Learning Framework for Multistage Parkinson's Disease Classification Using Acoustic Features of Sustained Korean Vowels." *Bioengineering, 10(8), 984.* (ISSN: 2306-5354, I.F. = 3.8) DOI: 10.3390/bioengineering10080984.
3. **Mondol, S.**, Kim, H. J., Kim, K. S., Lee, S. (2022). "Machine Learning-Based Hearing Aid Fitting Personalization Using Clinical Fitting Data." *J. Healthc. Eng., 2022, 1667672.* (ISSN: 2040-2309, I.F. = 3.05) DOI: 10.1155/2022/1667672.
4. Jeong, J., **Mondol, S.**, Kim, Y. W., Lee, S. (2021). "An Effective Learning Method for Automatic Speech Recognition in Korean CI Patients' Speech." *Electronics, 10(7), 807.* (ISSN: 2079-9292, I.F. = 2.6) DOI: 10.3390/electronics10070807.
5. **Mondol, S.**, Lee, S. (2019). "A Machine Learning Approach to Fitting Prescription for Hearing Aids." *Electronics, 8(7), 736.* (ISSN: 2079-9292, I.F. = 2.6) DOI: 10.3390/electronics8070736.

Full journal publication list available at: `simmraton.github.io`

Conferences

1. **Mondol, S.**, Yang, Y. J., Cho, J. R., Maeng, S., Lee, S. (2025). "Comparison of Key Acoustic Features of Voices of Patients with Depression and Parkinson's Disease." In *Proceedings of the 2025 Summer Conference of the Korean Society of Medical and Biological Engineering*, Jeju, Korea.
2. **Mondol, S.**, Jeon, J. H., Cho, J. R., Lee, S. (2024). "Stage-Specific Performance of Whisper ASR Model for Parkinson's Speech." In *Proceedings of the 2024 Fall Conference of the Korean Society of Medical and Biological Engineering*, Seoul, Korea.
3. **Mondol, S.**, Kim, R., Lee, S. (2023). "Data Valuation for the Small-Scale Parkinson's Speech Data Using Shapley Value." In *Proceedings of the 2023 Fall Conference of the Korean Society of Medical and Biological Engineering*, Seoul, Korea.
4. Bak, H., Cho, W., Kim, Y., **Mondol, S.**, Jeong, J., Kim, R., Lee, S. (2022). "Speech Recognition on Parkinson's Patients Based on Wav2Vec 2.0." In *Proceedings of the 2022 Summer Conference of the Korean Institute of Electrical and Electronic Engineers*, Jeju, Korea.
5. Bak, H., Cho, W., Kim, Y., **Mondol, S.**, Lee, S. (2022). "Conformer-CTC Based Korean Speech Recognition." In *Proceedings of the 2022 Winter Conference of the Korean Institute of Communications and Information Sciences (KICS)*, Pyeongchang, Korea.

Full conference publication list available at: `simmraton.github.io`

Outreach & Professional Development

Competition Question Developer & Reviewer Trainer *2021 – 2023*
Global IT Challenge for Youth with Disabilities, Korea

Team Leader *Jan. 2022*
BOLD Impact Prize in Entrepreneurship 2022, Inha University and BOLD School

Awards & Fellowships

Outstanding Research Award *2024*
Graduate School, Inha University

Jungseok International Scholarship (100%) *2017*
Graduate School, Inha University

Excellent Masters Thesis Award *2016*
International School, Chongqing University of Posts and Telecommunications

Chinese Government Scholarship (100%) *2013*
Chinese Government through Bangladesh Ministry of Education

Inter-University ICT Quiz Competition (2nd Prize) *2009*
Daffodil International University

References

Dr. Sangmin Lee (Ph.D. Advisor)
Professor, Department of Electrical and Computer Engineering,
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Dr. Zhou Yi (M.Eng. Advisor)
Professor, School of Communication and Information Engineering,
Chongqing University of Posts and Telecommunications, Chongqing 400065, China
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Dr. A K M Fazlul Haque (B.Sc. Advisor)
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